MB501P – CELL BIOLOGY

Practical Exam Notes

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M.C.Qs (Multiple Choice Questions)

1. The father of microscopy is **ANTONIE VAN LEEUWENHOEK**.
2. At the end of meiosis, **4** daughter cells are produced.
3. Animal cell and plant cell is distinguished by the presence or absence of **CELL WALL**.
4. The organelle responsible for photosynthesis is **CHLOROPLAST**.
5. **MITOCHONDRIA** is called powerhouse of cell.
6. There are **TWO** main components of dye.
7. Which technique is used to stain cellular nuclei? **FUELGEN STAIN**
8. DNA in eukaryotic cells is surrounded by **NUCLEAR MEMBRANE**.
9. Microbes under electron microscope are measure in **NANO METER 10^-9**.
10. The purpose of fixation in specimen preparation for light microscope is **PRESERVATION**.
11. Primary objective of staining in light microscopy is **CONTRAST**.
12. The feature which distinguishes eukaryotic cell from prokaryotic cell is presence or absence of **NUCLEUS**.
13. Which staining technique is commonly employed to highlight cellular nuclei in histology? **HEMATOXYLIN**
14. Result of cytokinesis in meiosis is **FOUR HAPLOID CELLS**.
15. Stage of mitosis in which nuclear membrane reforms is **TELOPHASE**.
16. The main objective to study Tradescantia stamina hair is **TO STUDY CYCLOSIS**.
17. The stains used to study the cell size and morphology in prokaryotes is **CRYSTAL VIOLET**.
18. Which of the following organelles is responsible for breaking down cellular waste in animal cells? **LYSOSOMES**.
19. What is the function of ribosomes in a prokaryotic cell? **PROTEIN SYNTHESIS**.
20. Smallest known bacteria is called **MYCOPLASMA**.
21. In gram staining, **IODINE** is used as a mordant.

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SHORT QUESTIONS

1. Write composition and function of cell wall.

Composition of Cell Wall

The cell wall is a rigid outer covering surrounding the plasma membrane. In plants, it is mainly composed of cellulose, along with hemicellulose and pectin. In fungi, it consists mainly of chitin and glucans, while in bacteria it is primarily made of peptidoglycan (murein).

Functions of Cell Wall

The cell wall provides shape, strength, and mechanical support to the cell. It protects the cell from physical damage and pathogens, prevents osmotic bursting by maintaining rigidity, helps maintain the cell's shape, and allows the passage of water and small molecules through its pores.

2. How meiosis is different from mitosis?

Difference Between Meiosis and Mitosis

Meiosis	Mitosis
Occurs in reproductive (germ) cells.	Occurs in somatic (body) cells.
Involves two successive divisions.	Involves one division.
Produces four haploid (n) daughter cells.	Produces two diploid (2n) daughter cells.
Daughter cells are genetically different.	Daughter cells are genetically identical.
Crossing over occurs during Prophase I.	Crossing over does not occur.
Functions in gamete formation and genetic variation.	Functions in growth, repair, and asexual reproduction.

3. What is simple staining?

Simple Staining

Simple staining is a basic microbiological staining technique in which only one basic dye is used to stain bacterial cells. Common dyes include methylene blue, crystal violet, and safranin. The positively charged dye binds to the negatively charged bacterial cell surface, making the cells clearly visible under a light microscope. It is mainly used to determine the shape, size, and arrangement of bacterial cells and to improve the contrast between the cells and the background for easier observation.

4. Differentiate between mitosis and meiosis on the basis of their division and how much cells they produce?

Difference Between Mitosis and Meiosis

Mitosis	Meiosis
Involves one cell division.	Involves two successive cell divisions (Meiosis I and Meiosis II).
Produces two daughter cells.	Produces four daughter cells.
Daughter cells are diploid (2n).	Daughter cells are haploid (n).

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5. What is cytokinesis and why it is essential?

Cytokinesis

Cytokinesis is the final stage of cell division in which the cytoplasm divides to form two separate daughter cells after the completion of mitosis or meiosis. In animal cells, it occurs by the formation of a cleavage furrow, whereas in plant cells it takes place through the formation of a cell plate, which develops into a new cell wall.

Importance of Cytokinesis

Cytokinesis is essential because it completes the process of cell division by separating the daughter cells. It ensures the equal distribution of cytoplasm, organelles, and other cellular contents between the daughter cells. It also allows each new cell to function independently, which is necessary for growth, tissue repair, reproduction, and normal development.

6. What is a microscope?

Microscope

A microscope is a scientific instrument used to magnify and observe objects that are too small to be seen with the naked eye. It enables the detailed study of cells, tissues, microorganisms, and other microscopic structures. Microscopes are widely used in biology, microbiology, medicine, and research laboratories. Depending on the type of microscope, objects can be magnified from tens to millions of times, allowing scientists to examine their structure and organization in detail.

7. What role does cyclosis play in plant cells?

Role of Cyclosis in Plant Cells

Cyclosis, also known as cytoplasmic streaming, is the continuous movement of the cytoplasm within plant cells. It helps transport nutrients, water, organelles, and other cellular materials from one part of the cell to another. Cyclosis ensures the even distribution of substances throughout the cell, improves the efficiency of metabolic activities, and facilitates the movement of chloroplasts to positions where they can receive maximum light for photosynthesis.

8. Write features of prokaryotic cell.

Features of Prokaryotic Cell

- ***Small and unicellular:*** Prokaryotic cells are microscopic, simple in structure, and usually consist of a single cell.
- ***No true nucleus:*** Their genetic material is not enclosed by a nuclear membrane; it is present in the nucleoid region.
- ***Lack membrane-bound organelles:*** Organelles such as mitochondria, endoplasmic reticulum, Golgi apparatus, and chloroplasts are absent.
- ***Circular DNA:*** They contain a single circular chromosome and often possess plasmids, which carry extra genetic information.
- ***70S ribosomes:*** Protein synthesis takes place on small 70S ribosomes present in the cytoplasm.
- ***Cell wall:*** Most prokaryotes have a rigid cell wall made of peptidoglycan, which provides shape and protection.
- ***External structures:*** Many have capsules for protection, pili for attachment and genetic exchange, and flagella for movement.
- ***Asexual reproduction:*** They reproduce mainly by binary fission, a simple and rapid method of cell division.
- ***Examples:*** Bacteria and archaea are the two major groups of prokaryotic organisms.

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9. Write some features of Protozoa.

Features of Protozoa

- Protozoa are unicellular, eukaryotic microorganisms with a well-defined nucleus and membrane-bound organelles.
- They are microscopic and are commonly found in freshwater, marine environments, moist soil, and inside other organisms.
- Most protozoa lack a cell wall and are covered by a flexible plasma membrane or pellicle.
- They obtain food by ingestion or absorption and are generally heterotrophic.
- Movement occurs through pseudopodia, cilia, or flagella, depending on the species.
- They reproduce mainly by binary fission, although some also reproduce sexually.
- Some protozoa are free-living, while others are parasitic and can cause diseases such as malaria and amoebic dysentery.
- Examples include Amoeba, Paramecium, Euglena, and Plasmodium.

10. Differentiate between acidic dye and basic dye.

Difference Between Acidic Dye and Basic Dye

Acidic Dye	Basic Dye
Acidic dyes carry a negative charge (anionic).	Basic dyes carry a positive charge (cationic).
They stain the background rather than the bacterial cell.	They stain the bacterial cell directly.
They are used in negative staining techniques.	They are used in simple staining and Gram staining.
Examples include Nigrosin and India ink.	Examples include Crystal violet, Methylene blue, and Safranin.

11. Why onion tips are used to observe mitosis?

Onion root tips are used to observe mitosis because they contain meristematic tissue, where cells divide rapidly for root growth. A large number of cells are therefore present in different stages of mitosis, making them easy to study under a microscope. In addition, onion chromosomes are large, distinct, and stain easily, allowing the stages of mitosis to be observed clearly.

12. Write the significance of cell division.

Significance of Cell Division

- Cell division is essential for the growth and development of multicellular organisms by increasing the number of cells.
- It replaces worn-out, damaged, or dead cells, helping in tissue repair and healing.
- It enables asexual reproduction in many unicellular organisms and some multicellular organisms.
- Meiosis produces gametes (sperm and eggs) required for sexual reproduction.
- It ensures the accurate transmission of genetic material from parent cells to daughter cells.
- Meiosis also creates genetic variation, which is important for adaptation and evolution.

13. Write two important components of gram staining.

- ***Crystal violet:*** Primary stain that initially stains all bacterial cells purple.
- ***Safranin:*** Counterstain that stains Gram-negative bacteria pink/red after decolorization.

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14. Name different types of microscope.

Types of Microscope

- ***Simple microscope:*** Uses a single convex lens for low magnification. It is mainly used for basic observation of small objects.
- ***Compound light microscope:*** Uses multiple lenses (objective and eyepiece) and visible light to produce higher magnification. Commonly used in laboratories to study cells and tissues.
- ***Electron microscope:*** Uses a beam of electrons instead of light, providing very high magnification and resolution. It is used to observe ultrastructural details of cells.
- ***Transmission Electron Microscope (TEM):*** Produces detailed images of internal cell structures by passing electrons through a thin specimen.
- ***Scanning Electron Microscope (SEM):*** Gives 3D-like images of the surface of specimens by scanning with electrons.
- ***Stereo (dissecting) microscope:*** Provides a 3D view of larger, opaque specimens at low magnification, used in dissection work.
- ***Phase contrast microscope:*** Enhances contrast in transparent, unstained living cells to observe internal structures.
- ***Fluorescence microscope:*** Uses fluorescent dyes and UV light to study specific structures or molecules within cells.

15. Name regions of root tip in onion.

Regions of Onion Root Tip

The onion root tip is divided into four main regions:

- ***Root cap:*** Protective region at the tip that protects the delicate meristematic tissue from mechanical damage as the root pushes through the soil.
- ***Meristematic region:*** Located just behind the root cap; cells divide actively by mitosis, responsible for root growth.
- ***Elongation region:*** Cells stop dividing and increase in size, contributing to the lengthening of the root.
- ***Maturation (differentiation) region:*** Cells differentiate into specialized tissues such as xylem, phloem, and epidermis for specific functions.

16. Why we stain microbes?

Microbes are stained because most of them are colorless, transparent, and very small, making them difficult to see clearly under a light microscope. Staining increases contrast between the microorganism and the background, allowing better visibility of cellular structures. Staining also helps in identifying the shape, size, and arrangement of microbial cells, such as cocci, bacilli, or spirilla. In differential staining techniques like Gram staining, it further helps to distinguish between different types of bacteria based on differences in their cell wall structure. This is important for identification, classification, and diagnosis of microorganisms in microbiology and medical laboratories.

17. Write a note on differential staining.

Differential Staining

Differential staining is a microbiological technique in which two or more stains are used to distinguish between different types of microorganisms or cellular structures. Unlike simple staining, it helps to identify bacteria based on their chemical and structural differences, especially cell wall composition. A common example is Gram staining, where bacteria are divided into Gram-positive (purple) and Gram-negative (pink/red) based on the thickness of peptidoglycan in their cell wall. Another example is acid-fast staining, used to identify bacteria like *Mycobacterium*, which resist decolorization due to waxy cell walls. Differential staining is widely used in bacterial identification, classification, and medical diagnosis.

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18. Enlist types of staining.

Types of Staining

- ***Simple staining:*** Uses a single dye to stain all bacterial cells uniformly. It helps in observing basic features like shape, size, and arrangement of microorganisms.
- ***Differential staining:*** Uses two or more dyes to distinguish between different types of bacteria based on structural differences. Example: Gram staining and acid-fast staining.
- ***Negative staining:*** Uses acidic dyes that stain the background instead of the cells. It is useful for observing delicate or unstained cells without distortion.
- ***Structural (special) staining:*** Used to highlight specific cell structures such as capsules, flagella, spores, or nuclei for detailed study.
- ***Gram staining:*** A widely used differential technique that classifies bacteria into Gram-positive (purple) and Gram-negative (pink) based on cell wall composition.
- ***Acid-fast staining:*** Used to identify acid-fast bacteria like *Mycobacterium*, which have waxy cell walls and resist decolorization.

19. Define prophase of mitosis.

Prophase of Mitosis

Prophase is the first and longest stage of mitosis, during which major structural changes occur in the nucleus and cytoplasm to prepare the cell for chromosome separation. During this stage, the chromatin material condenses and becomes visible as distinct chromosomes, each consisting of two identical sister chromatids joined at the centromere. The nucleolus gradually disappears, and the nuclear membrane starts breaking down, allowing the chromosomes to move freely in the cytoplasm. At the same time, the centrosomes move toward opposite poles of the cell and begin forming spindle fibers, which will later attach to the chromosomes and help in their proper separation during metaphase and anaphase.

20. Define smooth rough endoplasmic reticulum and rough endoplasmic reticulum.

Rough Endoplasmic Reticulum (RER)

Rough endoplasmic reticulum is a type of ER that has ribosomes attached on its outer surface, which gives it a rough appearance under the microscope. It is mainly involved in the synthesis of proteins, especially those that are secreted from the cell, inserted into membranes, or sent to lysosomes. These proteins are often transported to the Golgi apparatus for further processing and packaging.

Smooth Endoplasmic Reticulum (SER)

Smooth endoplasmic reticulum lacks ribosomes, so its surface appears smooth. It is mainly involved in the synthesis of lipids such as phospholipids and steroids, and also plays a role in detoxification of drugs and harmful chemicals. In some cells, it also helps in carbohydrate metabolism and calcium ion storage (especially in muscle cells).

21. What is significance of eukaryotes?

Significance of Eukaryotes

Eukaryotes are organisms whose cells contain a true nucleus and membrane-bound organelles, making them more complex and efficient in cellular organization. They are significant because they form all plants, animals, fungi, and protists, which include most visible life forms on Earth. Their cellular complexity allows specialization of cells and tissues**,** leading to the development of multicellular organisms with advanced functions such as respiration, movement, and reproduction. Eukaryotic cells enable efficient metabolic processes due to the presence of organelles like mitochondria (energy production) and chloroplasts

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(photosynthesis in plants). They also support sexual reproduction, which increases genetic variation and helps in evolution and adaptation of species.

22. Define chloroplast.

Chloroplast

Chloroplast is a double-membrane bound organelle found in plant cells and algae that contains the pigment chlorophyll. It is the site of photosynthesis, where light energy is converted into chemical energy in the form of glucose using carbon dioxide and water, releasing oxygen as a by-product.

LONG QUESTIONS

1. Write salient features of cell wall and cell membrane.

Salient Features of Cell Wall and Cell Membrane

Cell Wall

- Non-living, rigid outer covering present in plants, fungi, and bacteria.
- Gives shape, strength, and structural support to the cell.
- Mainly composed of cellulose (plants), chitin (fungi), and peptidoglycan (bacteria).
- Freely permeable, allowing water and small molecules to pass easily.
- Protects the cell from mechanical damage and pathogen attack.
- Prevents osmotic bursting by maintaining turgidity in plant cells.
- Helps in maintaining cell shape and rigidity.

Cell Membrane (Plasma Membrane)

- Thin, living, flexible boundary present in all living cells.
- Made up of phospholipid bilayer with embedded proteins (fluid mosaic model).
- Selectively permeable, controls entry and exit of substances.
- Regulates transport processes such as diffusion, osmosis, and active transport.
- Maintains internal balance (homeostasis) of the cell.
- Involved in cell communication and recognition through membrane proteins.
- Provides mechanical support along with flexibility, unlike the rigid cell wall.

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2. What is mordant in gram staining? Why it is used?

Mordant in Gram Staining

A mordant is a chemical substance used during staining procedures to enhance the intensity and permanence of a dye by forming an insoluble complex with it. In Gram staining, the mordant used is Gram's iodine solution.

Role and Function of Mordant (Gram's iodine)

- It reacts with the primary stain (crystal violet) inside bacterial cells.
- Forms a strong crystal violet–iodine (CV–I) complex, which is larger and insoluble in water.
- This complex becomes trapped within the thick peptidoglycan layer of Gram-positive bacteria.
- It increases the affinity of dye for the cell wall, making staining more stable.

Importance in Gram Staining

- Helps in differentiation of bacteria into Gram-positive and Gram-negative groups.
- During decolorization (alcohol/acetone wash), the CV–I complex is retained in Gram-positive bacteria due to thick peptidoglycan.
- In Gram-negative bacteria, the outer membrane is dissolved and the complex is lost, allowing them to take counterstain (safranin).
- Without mordant, the primary stain would be easily washed out, and proper differentiation would not occur.

3. Write similarities and dissimilarities between animal cell and plant cell.

SIMILARITIES & DISSIMILARITIES

Plant and animal cells are similar in many ways because both are eukaryotic cells. They possess a well-defined nucleus, cell membrane, cytoplasm, mitochondria, endoplasmic reticulum, Golgi apparatus, ribosomes, and DNA as genetic material. Both types of cells perform essential functions such as respiration, protein synthesis, growth, and metabolism, which are necessary for the survival of the organism.

Despite these similarities, plant and animal cells differ in several structural and functional features. Plant cells have a rigid cell wall made of cellulose, chloroplasts for photosynthesis, and a large central vacuole that maintains turgidity, while animal cells lack a cell wall and chloroplasts and usually have small, temporary vacuoles. Animal cells contain centrioles and more active lysosomes, which are generally absent in plant cells. Due to the presence of a cell wall, plant cells have a fixed, regular shape, whereas animal cells are more flexible and irregular in shape.

4. Explain heat fixation and chemical fixation method.

Heat Fixation

Heat fixation is a microbiological technique used to firmly attach bacterial cells onto a glass slide before staining. In this method, a thin smear of the culture is first prepared on a clean slide and allowed to air dry completely. After drying, the slide is passed through a flame or gently heated for a short time. This process kills the bacteria, preserves their general shape, and helps them stick to the slide so they are not washed away during staining. However, if excessive heat is applied, it may distort the cell structure.

Chemical Fixation

Chemical fixation is a method in which chemicals are used to preserve and attach cells or tissues instead of heat. Fixatives such as formaldehyde, ethanol, or glutaraldehyde are applied to the specimen to coagulate proteins and stop enzymatic activity. This preserves the internal and external structures of the cells more effectively than heat fixation. It is especially useful for delicate samples and tissues where heat could cause damage or distortion.

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5. Write characteristics of eukaryotic cell.

Characteristics of Eukaryotic Cell

- Eukaryotic cells have a true, well-defined nucleus surrounded by a double-layered nuclear membrane, which separates genetic material from the cytoplasm and controls gene expression.
- They contain membrane-bound organelles such as mitochondria, Golgi apparatus, endoplasmic reticulum, lysosomes, vacuoles, and (in plants) chloroplasts, each performing specific functions.
- Their genetic material is organized into multiple linear chromosomes associated with histone proteins, unlike circular DNA in prokaryotes.
- Eukaryotic cells are generally larger (10–100 µm) and more complex in structure compared to prokaryotic cells.
- The cytoplasm is highly compartmentalized, meaning different organelles perform specialized metabolic activities efficiently.
- They possess a well-developed cytoskeleton (microtubules, microfilaments, intermediate filaments) that maintains cell shape, supports organelles, and helps in movement.
- Ribosomes are of 80S type in the cytoplasm, responsible for protein synthesis, while mitochondria and chloroplasts contain 70S ribosomes.
- Cell division occurs through mitosis (for growth and repair) and meiosis (for gamete formation), ensuring genetic continuity and variation.
- Eukaryotic cells show complex regulation of gene expression, allowing cell specialization and formation of tissues and organs in multicellular organisms.

6. Explain mitosis with all phases.

Mitosis (Cell Division) with All Phases

Mitosis is a type of equational cell division in which one parent cell divides to form two genetically identical daughter cells, each having the same number of chromosomes as the parent cell. It is important for growth, repair, and replacement of damaged cells in organisms.

Interphase (Preparation stage – before mitosis)

- Cell is not actively dividing but prepares for division.
- DNA replicates and each chromosome forms two sister chromatids.
- Cell increases in size and synthesizes proteins and organelles needed for division.

Prophase

- Chromatin condenses into visible chromosomes (each with two sister chromatids).
- Nuclear membrane and nucleolus disappear.
- Spindle fibers begin to form from centrosomes and move to opposite poles.

Metaphase

- Chromosomes align at the equatorial (metaphase) plate of the cell.
- Spindle fibers attach to the centromere (kinetochore) of each chromosome.
- This ensures equal distribution of chromosomes.

Anaphase

- **Sister** chromatids separate at the centromere.
- Chromatids (now called daughter chromosomes) move toward opposite poles.
- Spindle fibers shorten, pulling chromosomes apart.

Telophase

- Chromosomes reach opposite poles and begin to decondense into chromatin.
- Nuclear membrane reforms around each set of chromosomes.
- Spindle fibers disappear.

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Cytokinesis (final stage)

- Division of cytoplasm occurs.
- In animal cells: cell membrane pinches in (cleavage furrow).
- In plant cells: cell plate forms, later developing into a new cell wall.
- Results in formation of two identical daughter cells.

7. Write typical structure and feature of bacterial cell. (Any Five)

Typical Structure and Features of Bacterial Cell

- ***Cell wall:*** Bacterial cell has a rigid cell wall mainly made up of peptidoglycan (murein) which provides shape, protection, and prevents osmotic bursting of the cell.
- ***Plasma membrane:*** A thin, selectively permeable membrane present beneath the cell wall that controls the entry and exit of substances and also plays a role in respiration in some bacteria.
- ***Nucleoid:*** Bacteria lack a true nucleus; instead, they have a nucleoid region containing a single circular DNA molecule that controls all cellular activities.
- ***Cytoplasm:*** A semi-fluid medium containing enzymes, nutrients, and ribosomes where most metabolic reactions take place.
- ***Ribosomes:*** Present as 70S type ribosomes, responsible for protein synthesis in bacterial cells.
- ***Capsule:*** Some bacteria have an outer gelatinous layer called capsule that provides protection, adhesion, and resistance against host defenses**.**
- ***Flagella:*** Some bacteria possess flagella which are long whip-like structures used for movement (motility)**.**

8. Write name and functions of reagents used in gram staining.

Reagents Used in Gram Staining and Their Functions

Crystal violet (Primary stain): It is the main dye that initially stains all bacterial cells purple, allowing visualization of both Gram-positive and Gram-negative bacteria.

Gram's iodine (Mordant): It forms a crystal violet–iodine (CV-I) complex with the primary stain, making it larger and more insoluble so that it gets strongly retained in Gram-positive cells.

Alcohol / Acetone (Decolorizer): It removes the stain from Gram-negative bacteria by dissolving the outer lipid membrane and washing away the CV-I complex, while Gram-positive bacteria retain the stain due to thick peptidoglycan.

Safranin (Counterstain): It stains the decolorized Gram-negative bacteria pink/red, providing contrast against Gram-positive bacteria, which remain purple.

9. Differentiate between functions of mitochondria and plastids.

Difference between Functions of Mitochondria and Plastids

Mitochondria are known as the "powerhouse of the cell" because they are responsible for cellular respiration. They convert glucose into ATP (energy currency of the cell) through aerobic respiration. This energy is used to perform various cellular activities such as growth, transport, and synthesis processes. Mitochondria are present in almost all eukaryotic cells (plants and animals) and are essential for energy production.

In contrast, plastids are mainly found in plant cells and are involved in synthesis and storage of food materials. Different types of plastids perform different functions: chloroplasts carry out photosynthesis by converting light energy into chemical energy (glucose), chromoplasts provide color to fruits and flowers, and leucoplasts store food such as starch, oils, and proteins. Thus, plastids are mainly associated with food production, storage, and pigmentation, while mitochondria are mainly responsible for energy production.

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10. Contrast between the cell wall of a plant cell and fungal cell, their composition and function.

Contrast Between Plant Cell Wall and Fungal Cell Wall

Basis	Plant Cell Wall	Fungal Cell Wall
Main composition	Mainly composed of cellulose microfibrils, hemicellulose, and pectin forming a strong polysaccharide network	Mainly composed of chitin (N-acetylglucosamine polymer) along with glucans and glycoproteins
Nature/structure	Rigid, thick, and highly organized structure providing fixed shape	Tough, flexible, and elastic structure allowing adaptation to environmental stress
Strength and support	Provides strong mechanical support and rigidity to plant tissues, helping plants stand upright	Provides structural strength and protection but allows some flexibility for growth and survival
Protection role	Protects against mechanical injury and excessive water uptake	Protects against desiccation, osmotic stress, and chemical damage
Osmotic regulation	Prevents cell bursting (lysis) by maintaining turgor pressure in hypotonic conditions	Maintains cell integrity under varying osmotic conditions, especially in soil or host environments
Functional significance	Helps in growth, shape maintenance, and tissue organization in plants	Helps in survival, pathogenicity (in some fungi), and adaptation to harsh environments
Additional feature	Often includes a middle lamella (pectin-rich layer) that binds adjacent plant cells	Lacks middle lamella; cell wall is more focused on individual cell protection and resilience

11. Explain Coarse Adjustment and Fine Adjustment.

Coarse Adjustment

Coarse adjustment is the larger focusing knob of the microscope that is used for initial focusing of the specimen. It moves the stage or body tube rapidly and in larger vertical steps**,** either upward or downward, to bring the specimen into approximate focus. It is mainly used under the low power objective lens because at this stage the risk of damage to the slide and objective is minimal. Coarse adjustment helps in quickly locating the specimen in the field of view, but it does not produce a sharp image. It should not be used under high power because it may cause damage to the slide or objective lens due to large movement.

Fine Adjustment

Fine adjustment is the smaller focusing knob of the microscope used after coarse focusing to obtain a clear and sharp image. It moves the stage or body tube very slowly and in very small increments, allowing precise focusing of fine cellular details. It is especially important under the high power and oil immersion objectives**,** where even slight movement can change clarity. Fine adjustment improves image sharpness, resolution, and detail without risking damage to the specimen or lens system.

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12. Write overview of Meiosis II.

Overview of Meiosis II

Meiosis II is the second division of meiosis, often called equational division because it resembles mitosis. It occurs in the two haploid cells produced after meiosis I and results in the formation of four genetically different haploid cells (gametes). Unlike meiosis I, there is no DNA replication before meiosis II, so each chromosome still has two sister chromatids.

Phases of Meiosis II

- ***Prophase II:*** Chromosomes condense again, nuclear membrane disappears (if reformed), and spindle fibers start forming in each haploid cell.
- ***Metaphase II:*** Chromosomes align individually at the equatorial plate, and spindle fibers attach to the centromeres from opposite poles.
- ***Anaphase II:*** The sister chromatids separate at the centromere and move toward opposite poles of the cell.
- ***Telophase II:*** Chromatids reach poles, decondense into chromatin, and nuclear membranes reform around each set.
- ***Cytokinesis:*** Cytoplasm divides, producing four haploid daughter cells, each genetically different.

13. Differentiate between Magnification power and resolution power of microscope.

Difference Between Magnification Power and Resolution Power of Microscope

Basis	Magnification Power	Resolution Power
Definition	Magnification power is the ability of a microscope to enlarge the image of an object so that it appears bigger than its actual size.	Resolution power is the ability of a microscope to distinguish two closely placed points as separate and clear images.
Nature	It is related to image size enlargement only.	It is related to image clarity and detail.
Dependence	Depends on the eyepiece and objective lenses magnification.	Depends on wavelength of light and numerical aperture of lenses.
Limitation	High magnification without good resolution gives a blurry image.	High resolution provides sharp and detailed image structure.
Importance	Helps in making small objects visibly larger.	Helps in observing fine cellular details and structures clearly.

14. Explain first stage of Mitosis.

First Stage of Mitosis (Prophase)

Prophase is the first and longest stage of mitosis, during which the cell prepares for chromosome separation. It begins after interphase and involves several important structural changes in both the nucleus and cytoplasm.

During prophase, the chromatin fibers condense and shorten to form visible chromosomes, each consisting of two identical sister chromatids joined at the centromere. The nucleolus gradually disappears, and the nuclear membrane starts breaking down, allowing the chromosomes to move freely within the cell. At the same time, the centrosomes migrate to opposite poles of the cell and begin organizing spindle fibers. These spindle fibers will later attach to the chromosomes and ensure their accurate separation during the next stages of mitosis. By the end of prophase, the cell is fully prepared for metaphase, where chromosomes align at the equatorial plate.

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15. Differentiate between cocci and bacillus with example.

Difference Between Cocci and Bacillus

Basis	Cocci	Bacillus
Shape	Spherical or round-shaped bacteria	Rod-shaped or cylindrical bacteria
Arrangement	May occur singly, in pairs (diplococci), chains (streptococci), clusters (staphylococci), or groups of four (tetrads)	Usually occur singly, in pairs (diplobacilli), or in chains (streptobacilli)
Occurrence	Commonly found on the skin, in the respiratory tract, and other body tissues	Commonly found in soil, water, food, and the intestinal tract
Motility	Most cocci are non-motile	Many bacilli are motile due to the presence of flagella, although some are non-motile
Examples	*Staphylococcus aureus, Streptococcus pyogenes, Neisseria gonorrhoeae*	*Escherichia coli, Bacillus subtilis, Lactobacillus acidophilus*

16. Write the main functions of Plastids in plant cells. Write a note on its types.

Plastids in Plant Cells

Plastids are **double membrane-bound organelles** found in plant cells and algae. They are responsible for the **manufacture, storage, and distribution of food**, as well as the production of pigments. Plastids possess their own DNA and ribosomes, allowing them to replicate independently.

Main Functions of Plastids

- Carry out **photosynthesis** and prepare food in the form of glucose.
- **Store food materials** such as starch, oils, and proteins.
- Produce and store **pigments** that give color to fruits, flowers, and leaves.
- Release **oxygen** as a by-product during photosynthesis.
- Help in the **growth, development, and metabolism** of plant cells.

Types of Plastids

1. Chloroplasts: Chloroplasts are green plastids containing the pigment chlorophyll. They are the site of photosynthesis, where sunlight, carbon dioxide, and water are converted into glucose and oxygen. They are mainly found in the green parts of plants, especially leaves.

2. Chromoplasts: Chromoplasts are colored plastids containing pigments such as carotene and xanthophyll**.** They provide yellow, orange, and red colors to flowers, fruits, and some roots. Their main function is to attract insects and animals for pollination and seed dispersal**.**

3. Leucoplasts: Leucoplasts are colorless plastids found in non-green parts of plants such as roots, tubers, and seeds. Their primary function is storage of food materials. Depending on the material stored, they are further classified into amyloplasts (store starch), elaioplasts (store oils), and proteinoplasts (store proteins).

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17. What is the most critical step while doing gram staining?

Most Critical Step in Gram Staining

The decolorization step is considered the most critical step in Gram staining because it determines whether bacteria will appear Gram-positive (purple) or Gram-negative (pink/red)**.** In this step, 95% ethanol or an acetone-alcohol solution is used as the decolorizing agent.

During decolorization, the alcohol dissolves the outer lipid membrane of Gram-negative bacteria, allowing the crystal violet–iodine (CV-I) complex to be washed out. In contrast, the thick peptidoglycan layer of Gram-positive bacteria retains the CV-I complex, so they remain purple.

If too much decolorizer is applied (over-decolorization), even Gram-positive bacteria may lose the primary stain and appear Gram-negative. On the other hand, if too little decolorizer is used (under-decolorization), Gram-negative bacteria may retain the purple stain and appear Gram-positive. Therefore, proper timing and careful application of the decolorizer are essential for obtaining accurate and reliable Gram staining results**.**

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